Wenqin Chen

Seattle, WA | wenqinchenphy@gmail.com | wenqin-chen.github.io | linkedin.com/in/wen-qin-chen

github.com/wenqin-chen | orcid.org/0000-0002-3445-7397

Research Interests

Machine learning for quantum many-body physics (neural quantum states); quantum geometry of superconductivity and phonon magnetism; topological phases, skyrmion crystals, and quantum anomalous Hall states; topological quantum computing with Majorana zero modes.

Education

University of Washington, Seattle, WA — 2021 – 2027 (expected)
PhD in Physics. Advisors: Di Xiao, Ting Cao

- Thesis: Machine Learning and Quantum Many-Body Theory for Condensed Matter Physics

Xi'an Jiaotong University, Xi'an, China — 2016 – 2021
BS in Physics (Honors)

- Outstanding Undergraduate Thesis Award

University of California, Berkeley, Berkeley, CA — 2019
Exchange student in Physics

Research Experience

University of Washington, Seattle, WA — 2021 – Present
Graduate Research Assistant. Advisors: Di Xiao, Ting Cao

- Neural quantum states: built a self-supervised transformer wavefunction ansatz that solves the continuum many-body Schrödinger equation in JAX, optimized by variational Monte Carlo with natural-gradient (stochastic reconfiguration / MinSR) updates, warm-started from Hartree-Fock and benchmarked against exact diagonalization (exact ground state matched within 0.7 mHa, 96% of the correlation energy, at two electrons); production runs at up to 18 electrons on H200 GPUs (UW Tillicum) and Google Cloud Platform.
- Open-sourced the solver as a reusable, CI-tested (GitHub Actions) NQS module with an exact-oracle pytest suite (attention ansatz, Metropolis MCMC sampler, Ewald Coulomb, MinSR optimizer): github.com/wenqin-chen/continuum-nqs.
- Quantum geometry of superconductivity: developed the Cooper-pair quadrupole framework showing Berry curvature and the quantum metric set a lower bound on the pair size in flat-band superconductors [arXiv:2605.03133]; applied to rhombohedral graphene, the predicted pair sizes match experimentally inferred coherence lengths.
- Phonon magnetism: built gauge-theory and geometric descriptions of giant phonon magnetic moments in Dirac materials [PRB 111, 035126; arXiv:2505.09732; arXiv:2605.06983], with predictions guiding magneto-Raman experiments by collaborators at Los Alamos (CINT) and UC Irvine.
- Co-developed the theory of adiabatic pumping of orbital magnetization by spin precession [PRL 134, 176702].

Los Alamos National Laboratory (T-4 / CNLS), Los Alamos, NM — 2023 – 2024
Research Intern (two summer internships). Mentor: Shi-Zeng Lin

- Delivered two first-author papers on phonon magnetism in consecutive summers [PRB 111, 035126; arXiv:2505.09732].
- Continuing the collaboration from UW, built a self-consistent Hartree-Fock framework to search for skyrmion crystals and quantum anomalous Hall order in strongly correlated two-dimensional electron systems; open-sourced the ~7k-line, 77-test solver core (github.com/wenqin-chen/hubbard-meanfield) and ran large-scale parameter-scan campaigns on the DOE NERSC Perlmutter supercomputer (SLURM, 128 cores/task).

Lawrence Berkeley National Laboratory, Berkeley, CA — 2019
Research Intern. Mentor: Joseph Orenstein

- Developed numerical simulations in direct collaboration with the experimental team to interpret optical birefringence measurements of the three-state nematic antiferromagnet Fe1/3NbS2; co-authored the resulting paper [Nature Materials (2020)].

Xi'an Jiaotong University, Xi'an, China — 2019 – 2021
Undergraduate Researcher. Mentor: Jie Liu

- Simulated non-Abelian braiding of Majorana zero modes in a tetron device and showed geometric-phase cancellation restores braiding-time-independent statistics in the presence of an Andreev bound state, the most common experimental contaminant in Majorana devices; first-author paper [PRB 105, 054507].
- Co-developed a minimal quantum-dot braiding protocol with electrical fermion-parity readout [SCPMA 64, 117811].

Publications

Nonadiabatic Theory of Phonon Magnetic Moments in Insulators and Metals — 2026
Haoran Chen, ***Wenqin Chen***, Kaijie Yang, Ting Cao, Di Xiao · *arXiv:2605.06983 (submitted to Physical Review Letters)*

Quantum Geometric Quadrupole of Cooper Pairs 2026
Wenqin Chen, Kaijie Yang, Ting Cao, Shi-Zeng Lin, Jiabin Yu, Di Xiao · *arXiv:2605.03133 (submitted to Physical Review Letters)*

Geometric Origin of Phonon Magnetic Moment in Dirac Materials 2025
Wenqin Chen, Xiao-Wei Zhang, Ting Cao, Shi-Zeng Lin, Di Xiao · *arXiv:2505.09732*

Adiabatic Pumping of Orbital Magnetization by Spin Precession 2025
Yafei Ren, ***Wenqin Chen***, Chong Wang, Ting Cao, Di Xiao · *Physical Review Letters 134, 176702*

Gauge Theory of Giant Phonon Magnetic Moment in Doped Dirac Semimetals 2025
Wenqin Chen, Xiao-Wei Zhang, Ying Su, Ting Cao, Di Xiao, Shi-Zeng Lin · *Physical Review B 111, 035126*

Non-Abelian Statistics of Majorana Zero Modes in the Presence of an Andreev Bound State 2022
Wenqin Chen, Jiachen Wang, Yijia Wu, Jie Liu, X. C. Xie · *Physical Review B 105, 054507*

Minimal Setup for Non-Abelian Braiding of Majorana Zero Modes 2021
Jie Liu, ***Wenqin Chen***, Ming Gong, Yijia Wu, X. C. Xie · *Science China Physics, Mechanics & Astronomy 64, 117811*

Observation of Three-State Nematicity in the Triangular Lattice Antiferromagnet Fe1/3NbS2 2020
Arielle Little, Changmin Lee, Caolan John, Spencer Doyle, Eran Maniv, Nityan L. Nair, ***Wenqin Chen***, Dylan Rees, Jörn W. F. Venderbos, Rafael Fernandes, James G. Analytis, Joseph Orenstein · *Nature Materials 19, 1062*

Talks and Presentations

Quantum Geometric Quadrupole of Cooper Pairs in Flat-Band Superconductors — APS Global Physics Summit, Denver *(contributed)*, Mar 2026

Quantum Geometric Quadrupole of Cooper Pairs in Flat-Band Superconductors — MEM·C IRG-2 Seminar, University of Washington *(invited seminar)*, Nov 2025

Chiral Phonons — poster, LANL–University of Michigan Joint Workshop on Quantum Materials, Los Alamos, NM, Jul 2025

Gauge and Geometric Origins of Phonon Magnetism in Dirac Materials — Oka Group Seminar, Institute for Solid State Physics, University of Tokyo, online *(invited seminar)*, May 2025

Geometric Theory of Phonon Magnetic Moment in Dirac Materials — APS Global Physics Summit, Anaheim *(contributed)*, Mar 2025

Phonon Magnetic Moments in Cd3As2: Theory Guiding the Magneto-Raman Experiment — presentations to the experimental collaboration, Los Alamos National Laboratory, online, Sep 2024

Theory of Giant Phonon Magnetic Moment in Doped Dirac Semimetals — APS March Meeting, Minneapolis *(contributed)*, Mar 2024

Dynamically Driven Orbital Magnetization Beyond the Adiabatic Limit — APS March Meeting, Las Vegas *(contributed)*, Mar 2023

Teaching

- Teaching Assistant, Department of Physics, University of Washington (Autumn 2021, Spring 2022)

Awards and Fellowships

- Clean Energy Institute Graduate Fellowship, University of Washington (2024–2025)
- Thouless Quantum Matter Graduate Fellowship, University of Washington
- Google Cloud Research Credits Award (2026)
- Outstanding Undergraduate Thesis Award, Xi'an Jiaotong University (2021)

Technical Skills

Machine learning: Deep learning (neural networks: transformer/attention architectures, neural quantum states), variational Monte Carlo, natural-gradient optimization (stochastic reconfiguration / MinSR), score-function (REINFORCE-style) gradient estimators, ablation studies, training stabilization (outlier-robust gradient clipping, learning-rate schedules, warm starts, early stopping)

Programming: Python (NumPy, SciPy, pandas, Matplotlib), JAX (Flax, Optax), PyTorch, Bash/Linux, Git, pytest

Scientific computing: Solid-state electronic-structure methods (exact diagonalization, self-consistent-field Hartree-Fock), Bogoliubov-de Gennes / Nambu modeling of superconductor-semiconductor (Majorana) nanowire devices, Berry-curvature and transport (Kubo) calculations, Markov chain Monte Carlo (Metropolis sampling), Ewald summation, numerical linear algebra

Infrastructure: GPU training in JAX (H200), SLURM/HPC job orchestration (NERSC Perlmutter, UW Tillicum), Google Cloud Platform, Azure, CI (GitHub Actions), reproducible research pipelines

Service and Activities

- Organizer, condensed-matter-theory group meetings, University of Washington (2025 – Present)
- Co-organizer, Summer Research Symposium for condensed matter physics, University of Washington (2025)
- Mentor, UStrive — virtual college-access mentoring for first-generation and low-income students (2025 – Present)
- Member, American Physical Society (Division of Condensed Matter Physics)